

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY

(Deemed University)

M.Tech/MS/PhD Examinations

Assessment – II - (Spring 2023)

Course Name: Data Science Tools & Techniques

Course Code: AM624D

Date: 23 – 03 - 2023

Max Marks: 10

Duration: 01 Hr

Note:

- Marks are indicated against each question. Write clearly and legibly Attempt all questions. Calculators are allowed.
- Before attempting, please ensure you have the correct question paper.

Q1	Derive Normal Equation for Linear Regression.	[2.5]
Q2	Suppose a sensor generates a sequence of numbers between 1 and 10 for each reading, WAP to compute the frequency of all values for each reading. Example: For six readings the array is: <pre> array([[9, 9, 4, 8, 8, 1, 5, 3, 6, 3], [3, 3, 2, 1, 9, 5, 1, 10, 7, 3], [5, 2, 6, 4, 5, 5, 4, 8, 2, 2], [8, 8, 1, 3, 10, 10, 4, 3, 6, 9], [2, 1, 8, 7, 3, 1, 9, 3, 6, 2], [9, 2, 6, 5, 3, 9, 4, 6, 1, 10]]) </pre> Then the output should be: <pre> [[1, 0, 2, 1, 1, 1, 0, 2, 2, 0], [2, 1, 3, 0, 1, 0, 1, 0, 1, 1], [0, 3, 0, 2, 3, 1, 0, 1, 0, 0], [1, 0, 2, 1, 0, 1, 0, 2, 1, 2], [2, 2, 2, 0, 0, 1, 1, 1, 1, 0], [1, 1, 1, 1, 1, 2, 0, 0, 2, 1]] </pre> <i>Handwritten notes:</i> 6x10 array index, 0 to 5 <i>Handwritten code:</i> ans = [[i.count(j) for j in range(1,11)] for i in arr]	[2.5]
Q3	Access the effects of outliers while training a linear regression with the help of an example.	[2.5]
Q4	How can you verify that gradient descent has converged at a global minimum for convex and non-convex error functions?	[2.5]

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ALL THE BEST

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arr[i] == 1

for i in range(0,6)

for i in range(1,11):
arr[0]